

A Sobolev Trace-Conjunction Bourgain–Brezis–Mironescu Formula

Automated Math Discovery Candidate Proof

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Status. Candidate full solution, likely valid pending expert review. For every $1 \leq p < \infty$, the source formula extends from compactly supported C^1 functions to the nonhomogeneous Sobolev space $W^{2,p}(\mathbb{R}_+^N)$. No optimality of this class is claimed.

Source and Open Problem

Let

$$\mathbb{H} = \mathbb{R}_+^N = \{(x', t) : x' \in \mathbb{R}^{N-1}, t > 0\}, \quad \partial\mathbb{H} = \mathbb{R}^{N-1} \times \{0\}.$$

Van Schaftingen [1] proves that if $u \in C_c^1(\overline{\mathbb{H}})$, then

$$\lim_{s \rightarrow 1^-} (1-s) \int_{\mathbb{R}^{N-1}} \int_{\mathbb{H}} \frac{|u(x', 0) - u(y)|^p}{|(x', 0) - y|^{N+sp}} dy dx' = c_{N,p} \int_{\mathbb{R}^{N-1}} |\nabla u(x', 0)|^p dx', \quad (1)$$

$$c_{N,p} := \frac{\pi^{(N-1)/2} \Gamma((p+1)/2)}{p \Gamma((N+p)/2)}.$$

Open Problem 1 on page 3 asks:

Does (1.8) still hold when u belongs to some suitable Sobolev space?

For trace conjunction integrals, we prove that if $u \in C_c^1(\bar{\mathbb{R}}_+^N)$, then [\(theorem 8.1\)](#)

$$(1.8) \quad \lim_{\substack{s \rightarrow 1 \\ <}} (1-s) \int_{\partial\mathbb{R}_+^N} \left(\int_{\mathbb{R}_+^N} \frac{|u(x) - u(y)|^p}{|x - y|^{N+sp}} dy \right) dx = \frac{\pi^{\frac{N-1}{2}} \Gamma(\frac{p+1}{2})}{p \Gamma(\frac{N+p}{2})} \int_{\partial\mathbb{R}_+^N} |Du|^p.$$

Compared to the classical Bourgain-Brezis-Mironescu formula [\(1.7\)](#), the right-hand side of its trace conjunction counterpart [\(1.8\)](#) features an integral that is only performed on the *boundary* but an integrand involving both the *tangential* and *normal* components of the derivative. The identity [1.8](#) is currently only proved for *smooth* functions.

Open Problem 1. Does [\(1.8\)](#) still hold when u belongs to some suitable Sobolev space?

For $v \in W^{1,p}(\mathbb{H})$, $1 \leq p < \infty$, we use the standard half-space trace $\text{Tr } v \in L^p(\mathbb{R}^{N-1})$, characterized by

$$v(\cdot, t) \longrightarrow \text{Tr } v \quad \text{in } L^p(\mathbb{R}^{N-1}) \quad (t \downarrow 0).$$

This characterization, including the endpoint $p = 1$, follows directly from absolute continuity on normal lines; see also [2, Chapter 18]. If $u \in W^{2,p}(\mathbb{H})$, both $\text{Tr } u$ and $\text{Tr } \nabla u$ are therefore well defined in L^p .

Proof Intuition

Write an interior point relative to a boundary point as $y = (x', 0) + r\omega$, where $r > 0$ and $\omega \in \mathbb{S}_+^{N-1} := \{\omega \in \mathbb{S}^{N-1} : \omega_N > 0\}$. The factor r^p in the first difference converts the energy into

$$(1-s) \int_{\mathbb{S}_+^{N-1}} \int_0^\infty r^{p(1-s)-1} \|Q_r^\omega\|_{L^p}^p dr d\omega, \quad Q_r^\omega = \frac{u(\cdot + r\omega', r\omega_N) - \text{Tr } u}{r}.$$

The measure $p(1-s)r^{p(1-s)-1} dr$ on $(0, 1)$ is a probability measure concentrating at $r = 0$.

The extra derivative in $W^{2,p}$ gives exactly the boundary regularity needed to identify what this Mellin average sees. Indeed, $\nabla u \in W^{1,p}$, so its horizontal slices converge strongly to $G = \text{Tr } \nabla u$. Integrating ∇u along the short ray and using translation continuity gives, uniformly in ω ,

$$Q_r^\omega \longrightarrow G \cdot \omega \quad \text{in } L^p.$$

The Mellin concentration now yields one factor $1/p$, while the hemisphere beta integral supplies the remaining gamma-function factor. Finally, the nonhomogeneous L^p norms of u and $\text{Tr } u$ control all radii $r \geq 1$, making that contribution $O(1-s)$.

Main Result

Theorem 1. *Let $N \geq 2$, $1 \leq p < \infty$, and let $u \in W^{2,p}(\mathbb{H})$ be real valued. Then*

$$\begin{aligned} \lim_{s \rightarrow 1^-} (1-s) \int_{\mathbb{R}^{N-1}} \int_{\mathbb{H}} \frac{|\text{Tr } u(x') - u(y)|^p}{|(x', 0) - y|^{N+sp}} dy dx' \\ = \frac{\pi^{(N-1)/2} \Gamma((p+1)/2)}{p \Gamma((N+p)/2)} \int_{\mathbb{R}^{N-1}} |\text{Tr } \nabla u(x')|^p dx'. \end{aligned} \tag{2}$$

Thus formula (1.8) of the source paper holds on the suitable Sobolev space $W^{2,p}(\mathbb{H})$, affirmatively answering Open Problem 1.

Uniform Boundary Expansion

Lemma 1. *Put $G = \text{Tr } \nabla u$. For $r > 0$ and $\omega \in \mathbb{S}_+^{N-1}$, define*

$$Q_r^\omega(x') = \frac{u(x' + r\omega', r\omega_N) - \text{Tr } u(x')}{r}.$$

Then

$$\lim_{r \downarrow 0} \sup_{\omega \in \mathbb{S}_+^{N-1}} \|Q_r^\omega - G \cdot \omega\|_{L^p(\mathbb{R}^{N-1})} = 0.$$

In particular, $\sup_{0 < r < r_0, \omega \in \mathbb{S}_+^{N-1}} \|Q_r^\omega\|_p < \infty$ for some $r_0 > 0$.

Proof. Set $F = \nabla u \in W^{1,p}(\mathbb{H}; \mathbb{R}^N)$. Strong convergence of horizontal slices to the trace gives

$$\eta(r) := \sup_{0 < t \leq r} \|F(\cdot, t) - G\|_p \longrightarrow 0.$$

Continuity of translations in $L^p(\mathbb{R}^{N-1})$ gives

$$\tau(r) := \sup_{|z| \leq r} \|G(\cdot + z) - G\|_p \longrightarrow 0.$$

The L^p -valued fundamental theorem of calculus along an oblique ray gives, for almost every $\omega \in \mathbb{S}_+^{N-1}$,

$$Q_r^\omega = \int_0^1 F(\cdot + tr\omega', tr\omega_N) \cdot \omega \, dt \quad \text{in } L^p(\mathbb{R}^{N-1}). \quad (3)$$

For completeness, one may first integrate from ε to r for a smooth approximation to u , pass to $W^{1,p}$, and then let $\varepsilon \downarrow 0$. The initial slice converges to $\text{Tr } u$; absolute continuity of the L^p -integral of ∇u justifies the last limit. This argument is valid for $p = 1$ as well.

For $0 < t \leq 1$, invariance of the L^p -norm under horizontal translations yields

$$\begin{aligned} & \|F(\cdot + tr\omega', tr\omega_N) - G\|_p \\ & \leq \|F(\cdot, tr\omega_N) - G\|_p + \|G(\cdot + tr\omega') - G\|_p \leq \eta(r) + \tau(r). \end{aligned}$$

Minkowski's integral inequality in (3) therefore gives

$$\|Q_r^\omega - G \cdot \omega\|_p \leq \eta(r) + \tau(r),$$

independently of ω . This proves both assertions. $\square$

Lemma 2 (Mellin concentration). *If $A : (0, 1) \rightarrow [0, \infty)$ is bounded and $A(r) \rightarrow A_0$ as $r \downarrow 0$, then*

$$\lim_{s \rightarrow 1^-} (1-s) \int_0^1 r^{p(1-s)-1} A(r) \, dr = \frac{A_0}{p}.$$

Proof. The measures

$$p(1-s)r^{p(1-s)-1} \mathbf{1}_{(0,1)}(r) \, dr$$

have total mass 1. For each fixed $\delta \in (0, 1)$, their mass on $[\delta, 1]$ is $1 - \delta^{p(1-s)}$, which tends to zero. Splitting the integral at δ , and then using $A(r) \rightarrow A_0$, proves the claim. $\square$

Proof of Theorem 1

Proof. Make the change of variables

$$y = (x', 0) + r\omega, \quad r > 0, \quad \omega \in \mathbb{S}_+^{N-1}.$$

Polar coordinates and Tonelli's theorem give, for $0 < s < 1$,

$$\begin{aligned} E_s(u) &:= (1-s) \int_{\mathbb{R}^{N-1}} \int_{\mathbb{H}} \frac{|\text{Tr } u(x') - u(y)|^p}{|(x', 0) - y|^{N+sp}} \, dy \, dx' \\ &= (1-s) \int_{\mathbb{S}_+^{N-1}} \int_0^\infty r^{p(1-s)-1} \|Q_r^\omega\|_p^p \, dr \, d\omega. \end{aligned} \quad (4)$$

Define

$$A(r) = \int_{\mathbb{S}_+^{N-1}} \|Q_r^\omega\|_p^p \, d\omega.$$

Lemma 1, the reverse triangle inequality, and continuity of $z \mapsto z^p$ show that A is bounded near zero and

$$A(r) \longrightarrow A_0 := \int_{\mathbb{S}_+^{N-1}} \|G \cdot \omega\|_p^p \, d\omega. \quad (5)$$

In fact A is bounded on all of $(0, 1)$. Lemma 1 handles small r ; away from zero, use

$$\|Q_r^\omega\|_p \leq r^{-1} (\|\mathrm{Tr} u\|_p + \|u(\cdot, r\omega_N)\|_p)$$

and the standard uniform slice bound $\sup_{0 < t \leq 1} \|u(\cdot, t)\|_p < \infty$. Lemma 2 therefore applies directly, and the part of (4) with $0 < r < 1$ converges to A_0/p .

We next compute A_0 . Since $\omega \mapsto |e \cdot \omega|^p$ is even, its integral over the upper hemisphere is one half of its integral over the whole sphere, regardless of the unit vector e . The beta integral gives

$$\int_{\mathbb{S}_+^{N-1}} |a \cdot \omega|^p d\omega = \frac{\pi^{(N-1)/2} \Gamma((p+1)/2)}{\Gamma((N+p)/2)} |a|^p \quad (a \in \mathbb{R}^N). \quad (6)$$

Indeed the left side for $|a| = 1$ is

$$\frac{1}{2} |\mathbb{S}^{N-2}| B\left(\frac{p+1}{2}, \frac{N-1}{2}\right).$$

By Fubini's theorem, (5) and (6) yield

$$\frac{A_0}{p} = \frac{\pi^{(N-1)/2} \Gamma((p+1)/2)}{p \Gamma((N+p)/2)} \int_{\mathbb{R}^{N-1}} |G(x')|^p dx'.$$

It remains to discard the part with $r \geq 1$. Returning to $h = r\omega$, the elementary inequality $|a - b|^p \leq 2^{p-1}(|a|^p + |b|^p)$ gives

$$\begin{aligned} & \int_{\substack{h_N > 0 \\ |h| \geq 1}} |h|^{-N-sp} \int_{\mathbb{R}^{N-1}} |\mathrm{Tr} u(x') - u(x' + h', h_N)|^p dx' dh \\ & \leq 2^{p-1} \|\mathrm{Tr} u\|_p^p \int_{\substack{h_N > 0 \\ |h| \geq 1}} |h|^{-N-sp} dh \\ & \quad + 2^{p-1} \int_0^\infty \|u(\cdot, t)\|_p^p \left[\int_{\substack{z \in \mathbb{R}^{N-1} \\ |(z, t)| \geq 1}} |(z, t)|^{-N-sp} dz \right] dt. \end{aligned} \quad (7)$$

For $s \in [1/2, 1)$, the first kernel integral is uniformly bounded. The bracketed integral is also bounded uniformly in $t > 0$ and such s . To see this, use $|(z, t)|^{-N-sp} \leq |(z, t)|^{-N-p/2}$ on $|(z, t)| \geq 1$. If $t \leq 1/2$, the restriction forces $|z| \geq \sqrt{3}/2$; if $t > 1/2$, integrate the latter kernel over all z and scale by t . Both bounds are uniform. Thus the right side of (7) is at most

$$C_{N,p} (\|\mathrm{Tr} u\|_p^p + \|u\|_{L^p(\mathbb{H})}^p).$$

After multiplication by $1 - s$, the $r \geq 1$ contribution tends to zero. Combining this with the small-radius limit proves (2). $\square$

Verification Notes

The accompanying script `code/verify_constants.py` performs two independent checks:

1. it compares the gamma-function expression in (6) with direct one-dimensional beta quadrature for several noninteger values of p and several dimensions;

2. it verifies the Mellin normalization on the test family $A(r) = A_0 + r^q$, for which the integral is exactly

$$\frac{A_0}{p} + \frac{1-s}{p(1-s)+q}.$$

These checks guard against a missing hemisphere factor or a missing factor p ; they are not used as proof.

The principal points for expert review are the L^p -valued ray identity (3), its uniformity as $\omega_N \downarrow 0$, and the uniform fixed-height kernel estimate in (7). Each is written so that the endpoint $p = 1$ remains visible.

Novelty Check, Scope, and Limitations

On July 23, 2026, bounded searches were run in arXiv and general web indexes for the exact source title, arXiv id 2502.18140, the quoted phrase “suitable Sobolev space,” and close combinations of “trace conjunction,” “Bourgain–Brezis–Mironescu,” and $W^{2,p}$. The searches found the source preprint and its published institutional repository record, but no later paper answering Open Problem 1 or proving the extension above. The run’s cheap indexes also had no prior result or attempt for this arXiv id before reservation. Novelty confidence is moderate pending a specialist citation search.

The theorem is a full answer to Open Problem 1 in its literal existence form: it supplies a suitable Sobolev space for every exponent covered by the smooth theorem. It does not claim that $W^{2,p}$ is sharp. A larger space characterized directly by an L^p boundary trace of the full gradient would also support the same proof. The result does not address the source paper’s separate converse problem concerning recovery of a weak normal derivative from finite limiting conjunction energy.

Human-Review Recommendation

Recommend expert verification as a candidate full solution. In particular, check:

1. the trace and oblique-ray fundamental theorem at $p = 1$;
2. the uniform estimate in Lemma 1;
3. the far-field estimate (7);
4. that the intended reading of “some suitable Sobolev space” accepts the concrete, non-sharp class $W^{2,p}$.

References

- [1] J. Van Schaftingen, *Trace conjunction inequalities*, Academy of Romanian Scientists. Annals. Series on Mathematics and its Applications **17** (2025), no. 1, 321–342, arXiv:2502.18140, doi:10.56082/annalsarsci.math.2025.1.321.
- [2] G. Leoni, *A First Course in Sobolev Spaces*, second edition, Graduate Studies in Mathematics 181, American Mathematical Society, Providence, RI, 2017.